

WEEK 8

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

Finish review

Question 1
Correct
Marked out of 1.00
Flag question

Coders here is a simple task for you, you have given an array of size N and an integer M .

Your task is to calculate the **difference between maximum sum and minimum sum of $N-M$ elements** of the given array.

Constraints:

$1 \leq t \leq 10$
 $1 \leq n \leq 1000$
 $1 \leq a[i] \leq 1000$

Input:

First line contains an integer T denoting the number of testcases.
First line of every testcase contains two integer N and M .
Next line contains N space separated integers denoting the elements of array

Output:

For every test case print your answer in new line

SAMPLE INPUT

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

```
1 #include<stdio.h>
2 int main()
3 {
4     int t;
5     scanf("%d",&t);
6     while(t-->0)
7     {
8         int n,m,d,min,temp;
9         scanf("%d %d",&n,&m);
10        d=n-m;
11        int arr[n];
12        for(int i=0;i<n;i++)
13            scanf("%d",&arr[i]);
14        for(int j=0;j<n;j++)
15        {
16            min=j;
17            for(int k=j;k<n;k++)
18            {
19                if(arr[k]<arr[min])
20                    min=k;
21            }
22            temp=arr[min];
23            arr[min]=arr[j];
24            arr[j]=temp;
25        }
26        int maxsum=0,minsum=0;
27        for(int a=0;a<d;a++)
28            minsum+=arr[a];
29        for(int b=n-1;b>=m-1;b--)
30            maxsum+=arr[b];
31        printf("%d\n",maxsum-minsum);
32    }
33 }
```

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

```
17         for (int k=j; k<N; k++)
18         {
19             if(arr[k]<arr[min])
20                 min=k;
21         }
22         temp=arr[min];
23         arr[min]=arr[j];
24         arr[j]=temp;
25     }
26     int maxsum=0,minsum=0;
27     for(int a=0;a<d;a++)
28         minsum+=arr[a];
29     for(int b=n-1;b>=m-1;b--)
30         maxsum+=arr[b];
31     printf("%d\n",maxsum-minsum);
32 }
33 }
```

	Input	Expected	Got	
✓	1	4	4	✓
	5 1			
	1 2 3 4 5			

Passed all tests! ✓

Question 2
Correct
Marked out of 1.00

A new deadly virus has infected large population of a planet. A brilliant scientist has discovered a new disease. Vaccine produced from this virus has various strength depending on midichlorians count. A person is cured only if midichlorians count in vaccine batch is more than midichlorians count of person. A doctor receives a new set of report of each infected patient. Practo stores all vaccine doctor has and their midichlorians count. You need to determine if doctor can save all

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

Search

ENG IN

21:57
12-01-2025

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

Question 2
Correct
Marked out of 1.00
Flag question

A new deadly virus has infected large population of a planet. A brilliant scientist has discovered a new strain of virus which can cure this disease. Vaccine produced from this virus has various strength depending on midichlorians count. A person is cured only if midichlorians count in vaccine batch is more than midichlorians count of person. A doctor receives a new set of report which contains midichlorians count of each infected patient, Practo stores all vaccine doctor has and their midichlorians count. You need to determine if doctor can save all patients with the vaccines he has. The number of vaccines and patients are equal.

Input Format

First line contains the number of vaccines - N. Second line contains N integers, which are strength of vaccines. Third line contains N integers, which are midichlorians count of patients.

Output Format

Print a single line containing 'Yes' or 'No'.

Input Constraint

$1 < N < 10$

Strength of vaccines and midichlorians count of patients fit in integer.

SAMPLE INPUT

5

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

Search

ENG IN

00:36
13-01-2025

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,min1,min2,temp,flag=1;
5     scanf("%d",&n);
6     int vac[n],pat[n];
7     for(int i=0;i<n;i++)
8         scanf("%d",&vac[i]);
9     for(int i=0;i<n;i++)
10        scanf("%d",&pat[i]);
11
12    for(int j=0;j<n-1;j++)
13    {
14        min1=j,min2=j;
15        for(int k=j;k<n;k++)
16        {
17            if(vac[k]<vac[min1])
18                min1=k;
19            if(pat[k]<pat[min2])
20                min2=k;
21        }
22        temp=vac[min1];
23        vac[min1]=vac[j];
24        vac[j]=temp;
25
26        temp=pat[min2];
27        pat[min2]=pat[j];
28        pat[j]=temp;
29    }
30    for(int i=0;i<n;i++)
31    {
32        if(vac[i]<=pat[i])
33        {
34            flag=0;
35            break;
36        }
37    }
38    if(flag==1
```

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

21:57
12-01-2025

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

```
24     vac[j]=temp;
25
26     temp=pat[min2];
27     pat[min2]=pat[j];
28     pat[j]=temp;
29 }
30 for(int i=0;i<n;i++)
31 {
32     if(vac[i]<=pat[i])
33     {
34         flag=0;
35         break;
36     }
37 }
38 if(flag==1)
39     printf("Yes");
40 else
41     printf("No");
42
43 }
```

	Input	Expected	Got	
✓	5	No	No	✓
	123 146 454 542 456			
	100 328 248 689 200			

Passed all tests! ✓

Question 3

21:57
12-01-2025

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

Question 3
Correct
Marked out of 1.00
Flag question

You are given an array of n integer numbers $a_1, a_2, \dots, a_n$. Calculate the number of pair of indices (i, j) such that $1 \leq i < j \leq n$ and $a_i \text{ xor } a_j = 0$.

Input format

- First line: n denoting the number of array elements
- Second line: n space separated integers $a_1, a_2, \dots, a_n$

Output format

Output the required number of pairs.

Constraints

- $1 \leq n \leq 10^6$
- $1 \leq a_i \leq 10^9$

SAMPLE INPUT

```
5
1 3 1 4 3
```

SAMPLE OUTPUT

```
2
```

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n, count=0;
5     scanf("%d", &n);
6     int arr[n];
7     for(int i=0; i<n; i++)
8         scanf("%d", &arr[i]);
9     for(int i=0; i<n-1; i++)
10     {
11         for(int j=i+1; j<n; j++)
12         {
13             if((arr[i]^arr[j])==0)
14                 count++;
15         }
16     }
17     printf("%d", count);
18 }
```

	Input	Expected	Got	
✓	5	2	2	✓
	1 3 1 4 3			

Passed all tests! ✓

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

Question 4
Correct
Marked out of 1.00
Flag question

You are given an array **A** of non-negative integers of size **m**. Your task is to sort the array in non-decreasing order and print out the original indices of the new sorted array.

Example:

A={4,5,3,7,1}

After sorting the new array becomes A={1,3,4,5,7}.

The required output should be "4 2 0 1 3"

INPUT :

The first line of input consists of the size of the array
The next line consists of the array of size m

OUTPUT :

Output consists of a single line of integers

CONSTRAINTS:

Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

Not secure | www.rajalakshmicolleges.org/moodle/mod/quiz/review.php?attempt=149975&cmid=187

REC-CIS

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     int arr[n];
7     for(int i=0;i<n;i++)
8         scanf("%d",&arr[i]);
9     int max=arr[0];
10    for(int i=1;i<n;i++)
11    {
12        if (arr[i]>max)
13            max=arr[i];
14    }
15    max++;
16    int min=0;
17    for(int a=0;a<n;a++)
18    {
19        for(int b=0;b<n;b++)
20        {
21            if(arr[b]<arr[min])
22                min=b;
23        }
24        printf("%d ",min);
25        arr[min]=max;
26    }
27 }
28 }
```

	Input	Expected	Got	
✓	5	4 2 0 1 3	4 2 0 1 3	✓
	4 5 3 7 1			